Dallas International School

Test 3

Topics 3.6 - 3.8 - 5.6 - 5.1

12th IB - Math AA

For SL and HL Students

EXERCISE 1 If $\cos x = -\frac{3}{4}$ and $\frac{\pi}{2} < x < \pi$, find $\tan x$ exactly.

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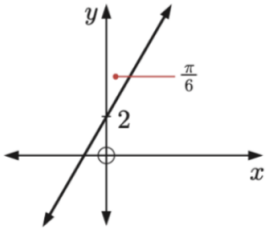
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EXERCISE 2 Find the equation of the line :



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EXERCISE 3 Find the least positive value of x for which $\cos\left(\frac{x}{2} + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$.

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EXERCISE 4 Solve $\sin^2 x + \sin x - 1 = 0.$, where $-\pi \leq x \leq \pi$.

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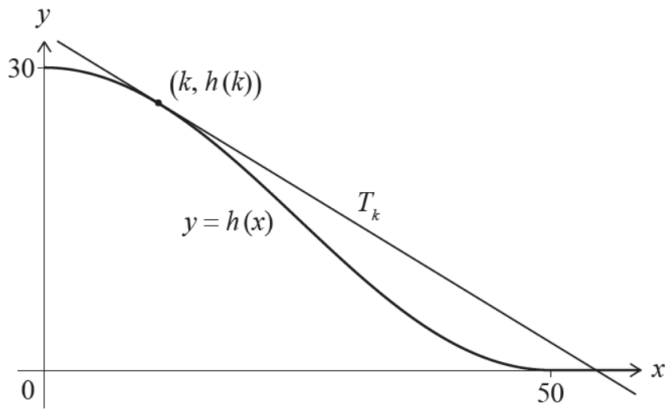
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EXERCISE 5 Consider the function $h(x) = 15\cos\left(\frac{\pi x}{50}\right) + 15,$ where $0 \leq x \leq 50.$

The tangent, T_k, to the curve $y = h(x)$ at the point $(k, h(k))$ is shown on the following diagram.



1. Find the gradient of T_k in terms of k.

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Consider the case where the angle between T_k and the x-axis is $\frac{\pi}{8}$ radians.

2. Find the possible values of k.

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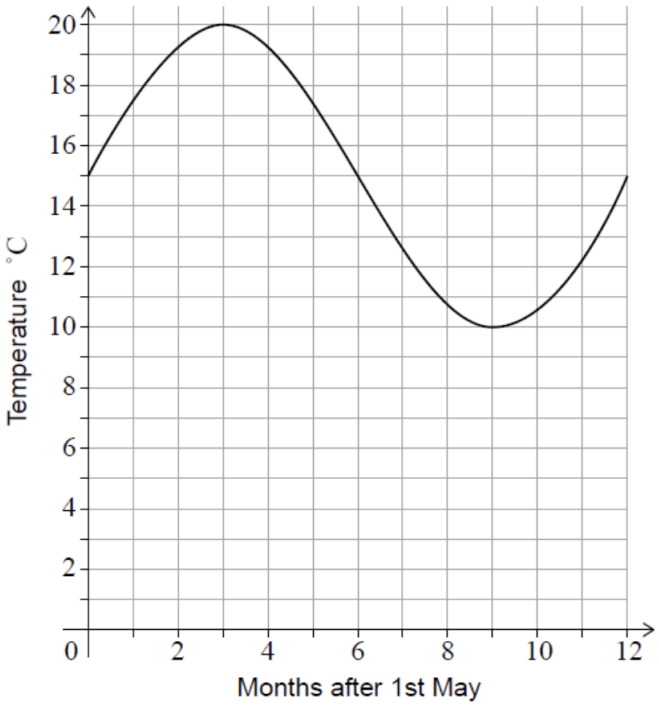
EXERCISE 6 Alex only swims in the sea if the water temperature is at least $15°C$. Alex goes into the sea close to home for the first time each year at the start of May, when the water becomes warm enough.

Alex models the water temperature at midday with the function

$$f(x) = a \sin \frac{\pi}{6} x + c$$

for $0 \leq x \leq 12$, where x is the number of months after 1st May and where $a, c > 0$.

The graph of $y = f(x)$ is shown in the following diagram.



1. Write down the value of : a and c.

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Alex is going on holiday and models the water temperature at midday in the sea at the holiday destination with the function $g(x) = 3.5 \sin \frac{\pi}{6}x + 11$, where $0 \leq x \leq 12$ and x is the number of months after 1st May.

2. Using this new model $g(x)$, find the midday water temperature on 1st October, five months after 1st May.

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3. **a.** Alex compares the two models and finds that $g(x) = 0.7f(x) + q$. Determine the value of q.

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b. BONUS : Show that the midday water temperature is never warm enough for Alex to swim.

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